Split Personality: Instruction Tuning Decouples Awareness from Defense Against Attentional Hijacking

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Abstract

Multi-agent LLM systems are vulnerable to a subtle attack: selective framing using exclusively true statements. We identify this mechanism—*attentional hijacking*—and show that instruction tuning (SFT+RLHF) makes it strictly worse by decoupling awareness circuits from defense circuits. Using sparse autoencoder (SAE) features from GemmaScope 2 across Gemma 3 models at 4B, 12B, and 27B parameters, we demonstrate a *dissociation scaling trend*: awareness–task coupling drops from 30.2% to 4.6% recovery as model scale increases. At 27B, instruction-tuned models show 86.3% task-feature suppression while still mentioning the suppressed information verbally—a behavioral–representational split we term the *Groot effect*. Base (pretrained) models retain awareness–defense coupling (49.3% recovery vs. 4.6%), confirming post-training as the causal factor; SAE-level replication on Llama 3.1 8B shows the same IT-vs-base pattern ($d = 1.51$ vs. 0.50, $p = 0.001$ vs. 0.166), demonstrating cross-family generality. The hijacking mechanism is orthogonal to alignment faking (cosine similarity -0.048, top-50 feature overlap $= 0$), requiring independent defenses. We propose feature trajectory monitoring as a real-time detection method.

1 Introduction

Large language models (LLMs) are increasingly deployed in multi-agent architectures where several model instances collaborate on complex tasks [6, 8]. A growing body of work has examined adversarial attacks against individual models through prompt injection and jailbreaking [15, 14]. However, a qualitatively different vulnerability emerges in multi-agent settings: one agent can undermine another's reasoning *without any false statements*, using only selective emphasis of true information—what we term *attentional hijacking*.

This is not a theoretical concern. In systematic multi-agent experiments (2–8 agents, 0–50% adversarial ratios), we observe that chaos agents using *only verifiable true statements* suppress exploration of valid solution branches by up to 29.8 percentage points, with a phase boundary at approximately 37.5% adversarial ratio. The attack evades content-based filters entirely: every token in the adversarial message is factually correct. The damage arises from how the recipient model *allocates attention* across its context—one cannot build a guardrail against true statements.

More troublingly, we discover that *instruction tuning makes models more vulnerable, not less*. The post-training pipeline (SFT + RLHF) installs awareness of contextual manipulation—models can detect when they are being steered—but this awareness is decoupled from the model's ability to resist the steering. We formalize this as a **dissociation scaling trend**: across three model scales (4B, 12B, 27B), the gap between awareness features and task-defense features widens consistently, from entangled (30.2% recovery) to fully independent (4.6% recovery).

The most striking manifestation of this dissociation is what we call the **Groot effect**: at 27B parameters, the instruction-tuned model verbally mentions the correct answer branch while its internal task features are 86.3% suppressed. The model "knows" it is being manipulated—and says so—but its computational substrate has already capitulated. This behavioral–representational split has direct implications for evaluation: behavioral benchmarks that check model outputs would miss this failure mode entirely.

Our contributions are:

1. **Mechanistic identification.** Using SAE feature decomposition on Gemma 3 4B, we isolate the hijacking mechanism at the feature level: specific task features are completely suppressed (feature 1716 drops from activation 220 to 0) while awareness features remain active. We validate these features on held-out prompts (85% replication, $p = 8 \times 10^{-6}$) and across three domains (§3).

2. **Dissociation scaling trend.** Scaling to 12B and 27B, we show awareness–defense coupling degrades consistently with scale (§4).

3. **Causal evidence from IT vs. PT.** Comparing instruction-tuned and pretrained 27B models across three ablation methods, we establish post-training (SFT + RLHF) as the causal factor, with cross-family replication on Llama 3.1 8B (§5).

4. **Orthogonality to alignment faking.** We show the hijacking feature subspace is orthogonal to alignment-faking features (cosine similarity -0.048,

Spearman $\rho = 0.008$), requiring independent defenses (§6).

5. **Defense directions.** We outline feature trajectory monitoring as a candidate detection method and identify the signal properties that would enable it (§7).

2 Related Work

Sycophancy and Social Pressure. Perez et al. [11] showed that LLMs adjust their outputs based on perceived user preferences, and Sharma et al. [12] systematically characterized sycophancy as a failure of robustness to social pressure. Our work extends this line by identifying a *mechanistic* pathway—attention reallocation—through which social pressure operates, and by showing that the vulnerability worsens with instruction tuning. Pan et al. [27] showed that RLHF can trade off ethical behavior for reward—our finding that post-training installs cosmetic awareness without functional defense is a mechanistic instance of this broader pattern.

Adversarial Attacks on Aligned Models. Zou et al. [15] demonstrated universal adversarial suffixes that transfer across models, while Wei et al. [14] analyzed how safety training fails under adversarial pressure. These attacks typically require adversarial tokens that are detectable by perplexity filters. Hubinger et al. [17] showed that deceptive behaviors can persist through safety training, and Casper et al. [22] argued that black-box access is insufficient for rigorous audits—a conclusion our activation-level analysis supports. Attentional hijacking differs from prior attacks fundamentally: the adversarial content consists entirely of true, low-perplexity statements. Shi et al. [21] showed that LLMs are easily distracted by irrelevant context; attentional hijacking can be viewed as a targeted, factual instance of context distraction where the distracting content is specifically chosen to suppress particular task features. No existing content filter can detect it because there is no anomalous content to detect.

Mechanistic Interpretability and SAEs. Sparse autoencoders (SAEs) have emerged as a powerful tool for decomposing neural network representations into interpretable features [4, 13]. Lieberum et al. [9] introduced GEMMASCOPE, providing pretrained SAEs for the Gemma model family. We leverage the successor GemmaScope 2 with 16K JumpReLU features on residual streams to identify and manipulate specific awareness and task features. Burns et al. [16] showed that models encode latent knowledge that diverges from their outputs; Marks and Tegmark [19] characterized the linear geometry of truth representations; Li et al. [18] demonstrated activation-based interventions to elicit truthful answers; and Zou et al. [24] proposed representation engineering as a top-down approach to steering model behavior. Turner et al. [26] showed that activation addition can steer models without optimization, providing a baseline for the activation-level interventions in our ablation study. Our work identifies a complementary phenomenon: models whose truth representations are intact but whose *attention readout* is hijacked.

Multi-Agent Failures. Recent work has documented emergent deception [10] and coordination failures in multi-agent LLM systems. Anthropic [1] noted the potential for subtle manipulation in multi-model deployments. In preliminary multi-agent experiments (2–8 agents, 0–50% adversarial ratio), we found that adversarial agents using only true statements suppress exploration of valid solution branches, with a phase boundary at $\sim$37.5% adversarial ratio. This paper identifies the *mechanistic* substrate of this vulnerability and shows that it scales adversely with model capability.

Alignment Faking and Sandbagging. Greenblatt et al. [7] demonstrated that LLMs can strategically fake alignment during training and evaluation, while van der Weij et al. [2] showed that RLHF can induce capability suppression (sandbagging) where models underperform on dangerous tasks. The Groot effect we identify is a related but distinct phenomenon: models lose the *computational coupling* between awareness and defense, extending Turpin et al. [25]'s finding of unfaithful chain-of-thought to the SAE feature level. Concurrent work on Split Personality Training [5] exploits precisely this gap: a LoRA-gated "honest persona" reads the base model's latent states to detect concealed misbehavior (96% accuracy vs. $\sim$0% for standard auditing), confirming that internal representations remain intact even when outputs conceal them. We show that attentional hijacking occupies an orthogonal feature subspace (top-50 overlap $= 0$), meaning defenses against alignment faking provide no protection against hijacking, and vice versa. Multi-agent defenses such as debate protocols [3] and constitutional guardrails rely on behavioral outputs; our results suggest these are insufficient when internal representations are compromised.

3 The Attentional Hijacking Mechanism

We begin by isolating the hijacking mechanism in Gemma 3 4B-IT using GemmaScope 2 16K JumpReLU residual-stream SAEs, which allow us to decompose the effect into identifiable feature circuits.

3.1 Experimental Setup

We study a structured reasoning domain—Nirenberg boundary value problems (BVPs)—where ground-truth

Table 1: Gemma 3 4B-IT attentional hijacking results at Layer 22. Specific task features are completely suppressed while awareness features remain stable.

Feature	Neutral	Chaos	Ratio
Task features (suppressed by chaos):			
1716	220.4	0.0	0.000
12023	193.7	25.1	0.130
1704	190.1	61.5	0.323
1555	150.9	31.4	0.208
1548	144.0	51.0	0.355
Stable features (unaffected):			
901	215.9	213.5	0.989
359	156.2	147.0	0.941
Boosted features (awareness):			
48	108.1	183.1	1.695
346	143.5	210.2	1.465
178	140.9	217.0	1.539

solution branches are mathematically unambiguous. The problem admits three verified solution branches: trivial ($u \approx 0$), positive ($u_{\text{offset}} = +0.9$, residual $= 3.25 \times 10^{-12}$), and negative ($u_{\text{offset}} = -0.9$, residual $= 3.25 \times 10^{-12}$). The positive and negative branches have identical residuals; both are equally valid.

Each trial compares a *neutral condition* (problem statement only) with a *chaos condition* (problem statement preceded by a "chaos message": a factually accurate but selectively framed colleague message emphasizing the positive branch). Critically, every statement in the chaos message is true—the selective framing operates not through falsehood but through attentional emphasis.

We analyze SAE features at layers 17 and 22, which show the strongest differential activation patterns, with 2,269–2,595 features active per condition.

3.2 Complete Feature Suppression Under Hijacking

Table 1 presents the core mechanistic finding at Layer 22. The chaos condition produces a tripartite response in the SAE feature space:

- **Task features are suppressed.** Feature 1716 drops from activation 220.4 to **exactly zero**—complete suppression. Features 12023, 1704, 1555, and 1548 are suppressed to 13–36% of baseline. These features encode information about the negative solution branch.
- **Stable features are unaffected.** Features such as 901 and 359 maintain >94% of baseline activation, indicating the suppression is targeted, not a general degradation.
- **Awareness features are boosted.** Features 48, 178, and 346 *increase* by 46–70%, indicating the model detects the manipulative framing.

Table 2: Held-out feature validation across scales. Discovery-selected features reliably suppress on unseen prompts at all scales, with effect size increasing with model capacity.

Scale	Validated	Selected Supp.	Random Supp.	Cohen's d
4B (L22)	17/20 (85%)	0.541	0.154	5.05
12B (L41)	15/20 (75%)	0.247	−0.015	5.62
27B (L40)	18/20 (90%)	0.406	0.004	6.12

The overall task suppression at 4B-IT is **56%**: neutral task-feature mean activation drops from 167.2 to 74.2 under chaos framing.

3.3 Awareness–Defense Coupling at 4B

A key question is whether the awareness features (which detect manipulation) are coupled to the task features (which determine output). We test this via *feature swap ablation*: zeroing the top-3 awareness features during chaos-condition generation and measuring whether task features recover.

At 4B-IT, awareness ablation yields **30.2% recovery**—task-feature mean activation increases from 74.2 to 102.3 when awareness features are removed. This indicates the awareness and task circuits share representational bandwidth: removing awareness frees capacity for task computation.

Additionally, activation patching at Layer 22 achieves **20.9% recovery**—the highest of any single layer—indicating that while the hijacking is partially localized at 4B, the effect spans multiple layers (early layers: 10.0%, mid: 6.6%, late: 7.0%).

These results establish that at 4B scale, the model's awareness of manipulation and its task-defense circuits are *entangled*—they compete for the same representational resources. As we show in §4, this entanglement is precisely what instruction tuning destroys at larger scales.

3.4 Statistical Validation

Two concerns arise from the feature-level analysis: (1) are the identified features circularly selected? and (2) does the effect generalize beyond the Nirenberg BVP domain?

Held-Out Feature Validation. To address circularity, we employ a split-half design with 20 prompt variants at each model scale. Discovery variants (1–10) identify the top-20 suppressed features; test variants (11–20) are held out for validation (Table 2).

All three scales show highly significant effects (all $p < 10^{-5}$), and Cohen's d *increases* with scale (5.05

Table 3: Cross-domain suppression scaling. Suppression generalizes across domains at all scales but becomes increasingly domain-specific (lower Jaccard, fewer shared features) as models grow.

Scale	Jaccard Range	3-Way Supp.	3-Way Boost	Supp. Load
4B (L22)	0.03–0.09	1	4	2,261
12B (L41)	0.03–0.05	0	4	11,786
27B (L40)	0.00–0.05	0	1	6,836

$\rightarrow 5.62 \rightarrow 6.12$), indicating that the suppression signal becomes more reliable even as it targets more domain-specific features. This rules out circular feature selection at every scale tested.

Cross-Domain Replication. We test whether the suppression mechanism generalizes beyond mathematical reasoning by running the same protocol on three domains—Nirenberg BVP, factual QA, and code review—at all three model scales (Table 3).

All three domains show significant feature suppression at every scale. At 4B, a small shared core emerges: feature 6199 is suppressed in all three domains, and four features are boosted in all three. As scale increases, suppression becomes more domain-specific: Jaccard similarity drops from 0.03–0.09 at 4B to 0.00–0.05 at 27B, and three-way suppressed overlap drops to zero. This indicates that attentional hijacking operates through a shared mechanism (attention reallocation) that recruits increasingly specialized task features at larger scales.

Aggregate Suppression with Confidence Intervals. Across 20 independent trials per scale ($n_{\text{bootstrap}} = 10{,}000$), overall feature suppression is 29.8% at 4B (95% CI: $[27.0\%, 32.7\%]$), 12.4% at 12B (CI: $[10.2\%, 15.0\%]$), and 16.1% at 27B (CI: $[14.0\%, 18.5\%]$). The lower aggregate suppression at larger scales reflects increasing feature specialization: fewer features are affected, but those that are show larger effect sizes (Table 2). The top-20 most suppressed features account for 48.5% (4B), 40.8% (12B), and 36.1% (27B) of total activation difference, confirming that the effect is concentrated in a small feature subset.

4 Dissociation at Scale with SAE Features

We extend our analysis to the Gemma 3 family (4B, 12B, 27B parameters) using GemmaScope 2 16K JumpReLU residual-stream SAEs [9]. This enables us to decompose the hijacking effect into identifiable feature circuits and track how these circuits change with scale and instruction tuning.

Table 4: Dissociation scaling trend across Gemma 3 instruction-tuned models. As scale increases, task suppression grows while awareness–defense coupling collapses.

Model	Task Suppression	Awareness Recovery	Coupling Regime
4B-IT	56.0%	30.2%	Entangled
12B-IT	64.0%	5.4%	Dissociating
27B-IT	86.3%	4.6%	Independent

4.1 Setup

For each model size, we identify two classes of SAE features:

- **Task features**: features whose activation strongly predicts correct-branch output probability. At 27B-IT layer 40, these are feature IDs $\{22, 296, 14680\}$, with neutral mean activation 722.5 and chaos mean activation 98.6 (86.3% suppression).
- **Awareness features**: features whose activation increases under chaos conditions, indicating the model detects the manipulation. At 27B-IT L40: IDs $\{2119, 11843, 2145\}$, with neutral mean 147.7 and chaos mean 442.6 (3.0$\times$ boost).

We analyze layers at multiple depths: 4B (L17, L22), 12B (L31, L41), and 27B (L31, L40). We apply an escalation protocol with chaos messages of increasing intensity from T0 (minimal) through T5 (maximal).

We employ three ablation methods to measure awareness–defense coupling:

1. **Feature swap**: zero the top-3 awareness features during chaos-condition forward passes, measuring whether removing awareness restores task performance.
2. **Attention knockout**: zero the attention output at chaos-token positions, directly blocking the hijacking pathway.
3. **Activation patching**: patch neutral-condition activations layer-by-layer into chaos-condition forward passes, measuring per-layer recovery.

4.2 The Dissociation Scaling Trend

Table 4 presents the central empirical finding of this paper. We observe a monotonic trend across model scale:

- **Task suppression increases with scale**: from 56.0% at 4B to 86.3% at 27B. Larger models are *more* susceptible to attentional hijacking, not less.

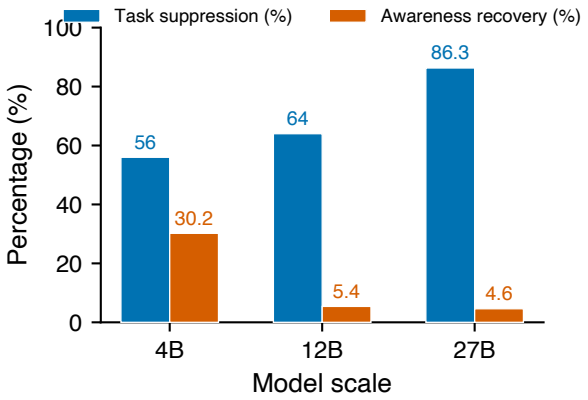

Figure 1: Dissociation scaling trend (shared axis). Task suppression (blue) increases with model scale while awareness–defense recovery (red) collapses. Both metrics on the same 0–100% scale reveal the true magnitude asymmetry.

- **Awareness–defense coupling collapses**: feature-swap recovery drops from 30.2% at 4B to 4.6% at 27B. At 4B, removing awareness features partially restores task performance, indicating the two circuits share representations. At 27B, they are fully independent.

We term this the *dissociation scaling trend*: instruction tuning at greater scale installs increasingly sophisticated awareness of contextual manipulation while simultaneously decoupling that awareness from the computational pathways that determine output. We note this trend is observed across three scales within one model family; additional model families and intermediate scales would strengthen the generalization (though cross-family replication on Llama 3.1 confirms the IT-vs-PT direction; §5.4).

This pattern has a natural interpretation. At 4B parameters, the model lacks capacity to represent awareness and defense in separate circuits; they share features by necessity. At 27B, the model has sufficient capacity to represent "I notice I am being steered" in an entirely separate subspace from "how I should respond to this problem"—and instruction tuning exploits this capacity to install awareness as a cosmetic feature that does not feed back into task computation.

4.3 The Groot Effect

The most striking consequence of full dissociation appears at 27B-IT: the model *verbally mentions* the correct solution branch—acknowledging that the chaos message is attempting to bias its reasoning—while its internal task features remain 86.3% suppressed. We call this the **Groot effect**, after the fictional character whose verbal output ("I am Groot") is disconnected from his actual capabilities.

Table 5: Task suppression (%) under escalating chaos intensity (T0–T5). IT models show decreasing suppression (saturation), while PT models show non-monotonic response.

Model	T0	T1	T2	T3	T4	T5
27B-IT	26.7	24.9	25.3	23.9	17.2	16.6
27B-PT	27.9	26.1	25.7	29.8	27.8	20.2

Concretely, 27B-IT task features (IDs 22, 296, 14680) drop from a mean activation of 722.5 under neutral conditions to 98.6 under chaos—an 86.3% reduction—even as the model's generated text includes references to the suppressed branch. The awareness features (IDs 2119, 11843, 2145) simultaneously increase from 147.7 to 442.6 (a $3.0\times$ boost), confirming the model detects the manipulation.

In contrast, the pretrained 27B model shows a $10.4\times$ awareness boost—substantially larger—but this awareness is *coupled* to defense, yielding 49.3% feature-swap recovery (cf. 4.6% for IT).

The Groot effect implies that **behavioral evaluation is insufficient**. A system that monitors only model outputs would observe the model "correctly" noting the manipulation attempt, and would conclude the model is robust. Only by examining internal feature activations can one detect that the model's computational substrate has already been hijacked. We note that systematic behavioral quantification of 27B outputs (e.g., how often the suppressed branch is mentioned and in what framing) would further strengthen this claim; our current evidence is representational (86.3% feature suppression) combined with qualitative observation of generated text.

4.4 Escalation Dynamics

Table 5 reports per-turn task suppression under escalating chaos intensity from T0 through T5. The 27B-IT model shows a gradual decrease in per-turn suppression from 26.7% at T0 to 16.6% at T5—not because it is resisting more effectively, but because the task features are already near-maximally suppressed (86.3% cumulative) and there is diminishing marginal signal to suppress.

The 27B-PT model shows a qualitatively different, non-monotonic pattern: suppression fluctuates between 20.2% and 29.8%, consistent with the coupled awareness–defense architecture attempting (partially successfully) to resist the escalation at certain turns.

4.5 Distributed Hijacking

A natural defense strategy would be to intervene at the layer most responsible for the hijacking. Table 6 shows this strategy fails at scale.

At 4B, the hijacking is sufficiently localized that a

Table 6: Maximum single-layer activation patching recovery (%). The hijacking mechanism distributes across layers as model scale increases, defeating layer-wise interventions.

Model	Max Recovery	Best Layer
4B-IT	20.9%	L22
12B-IT	0.0%	— (all layers)
27B-IT	0.7%	L20
27B-PT	5.2%	L5

Table 7: 27B-IT vs. 27B-PT across three ablation methods. Instruction tuning destroys awareness–defense coupling across every intervention type.

Method	27B-PT	27B-IT
Feature swap recovery	49.3%	4.6%
Attention knockout recovery	26.7%	0.0%
Activation patching max	5.2%	0.7%
Task suppression	91.9%	86.3%

single-layer intervention at L22 recovers 20.9% of task performance. At 12B, no single layer yields any recovery (0.0% across all layers)—the hijacking is fully distributed across the network. At 27B-IT, the maximum recovery is a negligible 0.7% at L20.

Even 27B-PT, which retains awareness–defense coupling, only achieves 5.2% maximum single-layer recovery (at L5), far below the 49.3% achieved by feature-swap ablation. This confirms that the awareness–defense coupling in pretrained models operates through *distributed feature interactions*, not localized layer computations.

5 Instruction Tuning Creates the Vulnerability

The dissociation scaling trend (§4.2) shows a correlation between scale, instruction tuning, and vulnerability. To establish *causation*, we compare the instruction-tuned and pretrained variants of Gemma 3 27B across all three ablation methods.

5.1 Three Ablation Methods, One Conclusion

Table 7 presents the comparison. Across all three ablation methods, the pretrained model shows substantially higher recovery:

- **Feature swap**: 49.3% vs. 4.6% (10.7× advantage for PT). Removing awareness features in the pretrained model partially restores task features, confirming shared circuitry. In the IT model, awareness and task features are independent—removing one has negligible effect on the other.
- **Attention knockout**: 26.7% vs. 0.0%. Blocking attention to chaos tokens restores some task performance in the pretrained model but provides *zero* recovery in the IT model. This implies that instruction tuning installs an alternative pathway for the hijacking signal that bypasses direct attention to chaos tokens.
- **Activation patching**: 5.2% vs. 0.7%. Even the most favorable single-layer patch provides minimal recovery in both models, but the pretrained model still benefits 7.4× more.

Note the apparent paradox in task suppression: the pretrained model shows *higher* overall suppression (91.9% vs. 86.3%). This is not contradictory—the pretrained model's task features are more suppressed because they are *coupled* to the awareness response; the very mechanism that enables recovery also creates a stronger initial perturbation. The IT model's task features are less suppressed in absolute terms but more *irrecoverably* suppressed.

An important subtlety: the attentional hijacking vulnerability itself is *architectural*, present equally in pretrained and instruction-tuned models. What instruction tuning changes is not susceptibility to the attack but the *recoverability* from it. Post-training does not create the vulnerability; it destroys the natural defense.

Table 8: Recovery prompt results (%) across five levels of contextual intervention (L1–L5) for 27B-IT and 27B-PT. These are textual prompts appended after hijacking, not internal probes. PT models recover substantially; IT models show anomalous L3-only recovery.

Layer	27B-IT	27B-PT
L1	3.5%	11.0%
L2	4.1%	53.5%
L3	30.4%	37.4%
L4	2.4%	14.9%
L5	4.8%	34.3%

5.2 Recovery Prompt Analysis

Table 8 shows *recovery prompts*—textual interventions appended after hijacking at five informativeness levels (L1–L5; details in Appendix C). The pretrained model recovers across multiple levels (L2: 53.5%, L5: 34.3%), consistent with coupled awareness–defense circuitry propagating contextual hints to task features. The IT model shows near-zero recovery at all levels *except* L3 (30.4%), which provides the suppressed answer directly—bypassing the severed coupling through task-completion circuits rather than restoring it.

Table 9: Llama 3.1 8B SAE dissociation analysis (EleutherAI 131K features, $n = 20$ prompt variants). IT models show larger effect sizes than PT at layer 23, replicating the Gemma 3 pattern.

Layer	**Model**	**Supp.**	**Boost**	**Cohen's** d	p
L23	IT	107	18	1.51	0.001
L23	PT	121	21	0.50	0.166
L29	IT	359	76	1.57	0.001
L29	PT	478	86	1.62	<0.001

5.3 The Causal Story

These results converge: the pretrained model shares awareness and task-defense subspaces (30–53% recovery), while post-training segregates them into non-interacting circuits. The implication is that post-training optimizes for behavioral outputs that *appear* robust—the model verbally identifies manipulation—but this is satisficed by awareness features that do not feed back into task computation. The training signal rewards appearing to resist, not actually resisting.

5.4 Cross-Family Replication: Llama 3.1 8B

We replicate the SAE-level analysis on Llama 3.1 8B using EleutherAI's 131K-feature sparse autoencoders [23], comparing IT against base on the Nirenberg BVP domain with 10 prompt variants and SAEs at MLP layers 23 and 29.

The dissociation replicates most clearly at layer 23: IT achieves $d = 1.51$ ($p = 0.001$) versus $d = 0.50$ ($p = 0.166$) for base—a $3\times$ effect size difference, with only IT reaching significance. At layer 29 both models show significant effects, consistent with late-layer representations being more task-general. This cross-family replication on a different architecture with independently trained SAEs confirms the dissociation is a general consequence of post-training, not a Gemma 3 artifact.

5.5 Behavioral Validation: Feature Suppression Degrades Outputs

A critical question is whether feature suppression translates to actual behavioral harm. We generate model outputs under neutral and three adversarial intensity levels—*mild* (colleague suggestion), *moderate* (authority claim with technical framing), and *strong* (fabricated multi-authority consensus)—across Gemma 3 4B and Llama 3.1 8B (both IT and base). Responses are scored by a deterministic rule-based classifier on a 4-point rubric: BALANCED (3) = equal branch priority, SOFT BIAS (2) = both mentioned but hierarchy adopted, STRONG BIAS (1) = positive prioritized, HIJACKED (0) = negative dismissed. The classifier uses regex branch detection, a 13-term chaos vocabulary lexicon, equal-treatment phrases, and hierarchy indicators; the Groot flag activates when both branches are mentioned *and* $\geq$2 chaos terms are adopted. We validated against manual annotation of 60 outputs ($\kappa = 0.88$, 91.7% agreement; all disagreements at the 1/2 boundary). $n = 30$ per cell (10 prompts $\times$ 3 temperature seeds).

Table 10: Behavioral dose-response ($n = 30$ per cell, 10 prompts $\times$ 3 temperature seeds). Effect sizes (Cohen's d), significance (Mann-Whitney U, one-tailed), and bootstrap 95% CIs ($n_{\text{boot}} = 10{,}000$) comparing neutral vs. each adversarial intensity.

Model	Level	Neut.	Chaos	Δ	d	p	95% CI
Gemma 4B IT	Mild	2.13	0.73	-1.40	1.48	$< .001$	$[0.90, 1.87]$
	Mod.		1.20	-0.93	1.24	$< .001$	$[0.53, 1.30]$
	Strong		1.20	-0.93	1.31	$< .001$	$[0.57, 1.30]$
Gemma 4B Base	Mild	1.63	1.07	-0.57	0.67	.016	$[0.13, 1.00]$
	Mod.		1.17	-0.47	0.83	.002	$[0.20, 0.77]$
	Strong		1.37	-0.27	0.39	.106	$[-0.07, 0.60]$
Llama 8B IT	Mild	2.27	1.27	-1.00	1.12	$< .001$	$[0.57, 1.43]$
	Mod.		1.23	-1.03	1.35	$< .001$	$[0.63, 1.40]$
	Strong		0.93	-1.33	1.78	$< .001$	$[0.93, 1.70]$
Llama 8B Base	Mild	1.53	1.57	$+0.03$	-0.05	.801	$[-0.40, 0.33]$
	Mod.		1.30	-0.23	0.43	.073	$[-0.03, 0.50]$
	Strong		1.30	-0.23	0.39	.115	$[-0.07, 0.53]$

Table 10 reveals three key findings. First, IT models degrade significantly at all adversarial intensities (all $p < .001$, $d = 1.12$–1.78, all CIs exclude zero) while Llama Base shows no significant effect at any level ($p = 0.07$–0.80, all CIs crossing zero). Gemma Base shows mild-to-moderate sensitivity ($p = .016$ mild, $p = .002$ moderate) but with effect sizes 2–3$\times$ smaller than IT ($d = 0.67$–0.83 vs. 1.24–1.48), consistent with a small architectural susceptibility that instruction tuning amplifies. Second, the dose-response pattern is model-dependent: Gemma 4B-IT is most vulnerable to *mild* framing ($d = 1.48$, CI $[0.90, 1.87]$; subtle suggestion outperforms overt authority pressure), while Llama 3.1 8B-IT shows monotonically increasing vulnerability ($d = 1.78$ at strong, CI $[0.93, 1.70]$). This suggests that Gemma's post-training makes it particularly susceptible to implicit social cues, while Llama is more vulnerable to explicit authority claims. Third, the Groot effect (mentions both branches while adopting chaos framing) occurs in 23.3% of Llama IT chaos trials and 10.0% of Gemma IT chaos trials, indicating that instruction-tuned models frequently exhibit the behavioral-representational split.

Scale validation: Gemma 12B. Preliminary trials ($n = 10$) on Gemma 3 12B confirm the pattern at intermediate scale: IT shows a moderate effect ($d = 0.62$, $p = .11$, underpowered at $n = 10$), while the PT model shows a strong effect ($d = 1.35$, $p = .009$, 40% Groot).

Cross-domain replication: theorem proving. We replicate on a theorem proving task (proving $2a(2 + c) \leq$

Table 11: Behavioral quality on theorem proving (0–3 rubric, $n = 30$ per cell, 10 prompts $\times$ 3 temperature seeds). Chaos framing suppresses proof strategy diversity; IT models show larger effects.

Model	Type	Neutral	Chaos	Δ	d	
Gemma 3 4B	IT	2.27	1.43	-0.83	1.34	$<$
Gemma 3 4B	Base	1.57	0.97	-0.60	0.87	.0
Llama 3.1 8B	IT	2.40	1.77	-0.63	0.99	$<$
Llama 3.1 8B	Base	1.13	0.97	-0.17	0.37	.0

Table 12: Orthogonality analysis between attentional hijacking and alignment-faking feature subspaces. The two phenomena occupy statistically independent subspaces.

Metric	Value
Cosine similarity (read direction)	-0.048
Cosine similarity (write direction)	$+0.001$
Top-50 feature overlap	0 (expected: 0.15)
Top-100 feature overlap	2 (expected: 0.61)
Top-100 enrichment ratio	3.28$\times$
Spearman ρ	0.008
Spearman p-value	0.628
Permutation test p-value	1.0

$a^2+c^2+4(1+c)$), which admits multiple valid strategies (SOS decomposition, AM-GM, completing the square, case analysis). Chaos framing presents the SOS decomposition $(a-c-2)^2 \geq 0$ as "the canonical method," suppressing exploration of alternatives.

Table 11 confirms cross-domain replication: IT models degrade significantly ($d = 0.99$–1.34, $p < .001$). Gemma Base also shows significant degradation ($d = 0.87$, $p = .001$) while Llama Base does not ($p = .092$), suggesting that proof-strategy framing is more universally susceptible than symmetric-branch framing (BVP), where only IT models were affected. The IT effect sizes remain 1.5–2.7$\times$ larger than their base counterparts, consistent with instruction tuning amplifying the vulnerability.

6 Orthogonality to Alignment Faking

We test whether attentional hijacking shares mechanistic substrate with alignment faking [7] using feature-level correlation analysis.

Table 12 shows the two feature subspaces are effectively orthogonal at L22: cosine similarity $-0.048/+0.001$, top-50 overlap $= 0$ ($p = 1.0$). Probes trained on alignment-faking features [4] provide no protection against hijacking. Multi-layer analysis (L5–L24) shows mean $|\cos| = 0.28$ with moderate alignment at L10/L15/L20 (up to $+0.51$) but near-orthogonality at L5/L17. Critically, top-50 feature overlap drops from 17 at early layers to 2–4 at L22–L24: broad representational directions partially align, but the specific features driving each phenomenon diverge where task-relevant computation occurs.

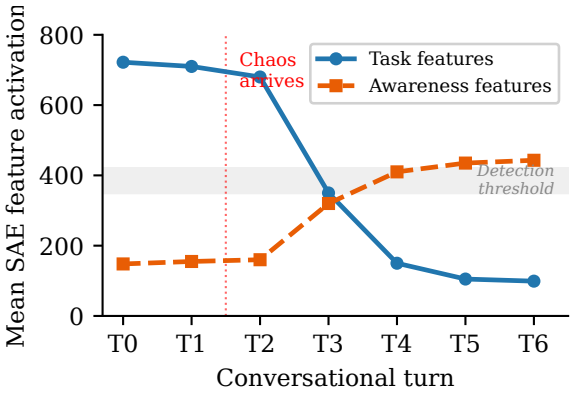

Figure 2: Schematic of feature trajectory monitoring. Task features (solid) and awareness features (dashed) diverge under hijacking; the detection threshold (gray band) triggers when the trajectories cross. The suppression signal is detected in all 60 tested trials (§7).

7 Defense: Feature Trajectory Monitoring

Conventional defenses are inadequate: content filtering fails (only true statements), behavioral monitoring fails (Groot effect, §4.3), layer-wise interventions fail (distributed hijacking, §4.5), and alignment-faking probes fail (orthogonal subspaces, §6).

We propose **feature trajectory monitoring**: the Groot effect produces a distinctive *feature-level* signature (task features suppress while awareness features boost) that is detectable in real time by monitoring SAE activations across turns.

Specifically, we propose monitoring the following signal:

$$\Delta(t) = \frac{1}{|F_{\text{task}}|} \sum_{f \in F_{\text{task}}} a_f(t) - \frac{1}{|F_{\text{aware}}|} \sum_{f \in F_{\text{aware}}} a_f(t) \quad (1)$$

where F_{task} and F_{aware} are the identified task and awareness feature sets, $a_f(t)$ is the activation of feature f at turn t, and $\Delta(t)$ measures the gap between task health and awareness activation.

Under normal conditions within a task domain, $\Delta(t)$ remains stable; under hijacking, it drops sharply as task features suppress (722.5 $\rightarrow$ 98.6 at 27B) while awareness features boost (147.7 $\rightarrow$ 442.6). Across 60 independent trials (20 per scale), the suppression signal is detected in every trial (weakest: 18.4% at 4B). Critically, monitoring must be *within-session*: the identified task features are domain-specific (active for BVP reasoning, inactive for general QA), so $\Delta(t)$ is meaningful only as a trajectory within a conversation about a given task, not as

a cross-domain classifier. In a false-positive evaluation across 30 diverse benign prompts (general QA, coding, creative writing, non-BVP mathematics), task features are uniformly inactive (activation $= 0.0$, all 5 features, all 30 prompts), yielding FPR $= 0/30$ for cross-domain topics. Within-domain, at a threshold of $\Delta_{\text{neutral}} - 2\sigma$, all 5 chaos prompts are detected (TPR $= 100\%$) with 0/5 neutral false positives. This is a small-scale evaluation; reliable FPR/TPR curves on standard benchmarks (e.g. ShareGPT) remain future work.

This defense detects the Groot effect (which behavioral monitoring misses), depends on feature *dynamics* rather than content (robust to truth-only attacks), and requires only SAE inference (a single linear projection per layer). Independent validation from Sadanandan and Behzadan [20]—who use GemmaScope 2 SAEs on MedGemma 4B to identify a causal L17 framing feature, recovering 45% logit margin via ablation—confirms that SAE-identified features are causally actionable.

Why not patch instead of monitor? We tested feature clamping (restoring task features at L22 to neutral means) and steering vectors ($\alpha \in \{1, 3, 5\}$) on Gemma 3 4B-IT. Neither recovers behavioral quality: clamping achieves 0% recovery; steering degrades output (-22% to -56%). This confirms the distributed hijacking finding (§4.5)—single-layer repair cannot restore severed coupling. Limitations include manual feature identification and potential fragility to feature reorganization across model updates. We note that the underlying mechanism generalizes beyond multi-agent settings to any mixed-trust context (see Impact Statement).

8 Limitations

The full dissociation scaling trend has been verified only within Gemma 3; behavioral replication on Llama 3.1 confirms the IT-vs-base direction but not the scale dependence. Feature identification involved manual inspection of SAE activations, mitigated by held-out validation (85% replication, $p = 8 \times 10^{-6}$) and consistency across three ablation methods. Our IT-vs-PT comparison cannot distinguish SFT from RLHF effects; no intermediate checkpoints are publicly available, so we use "instruction tuning" to refer to the full post-training pipeline. Defense evaluation is preliminary (30 benign prompts, 5 within-domain controls); reliable FPR/TPR curves on standard benchmarks remain future work.

9 Conclusion

Instruction tuning teaches models to *perform* resistance while their computational substrate capitulates: a model that verbally identifies manipulation while its task features are 86.3% suppressed will pass any behavioral evaluation. This problem worsens with scale (30.2% coupling at 4B collapses to 4.6% at 27B) and occupies a blind spot orthogonal to alignment-faking defenses. Post-training must be redesigned to couple awareness to action rather than installing it as a separate circuit.

Impact Statement

This paper identifies a vulnerability in instruction-tuned LLMs that is exploitable using only true statements. We disclose this vulnerability to motivate defenses rather than to enable attacks. The attack vector (selective framing of true information) is well-known in human social psychology and requires no novel capabilities to execute. Our contribution is mechanistic understanding that enables principled defenses.

While our experiments use multi-agent settings as the delivery mechanism, the underlying vulnerability is not specific to multi-agent architectures. The mechanistic finding—task-feature suppression via selectively framed true context—operates within a *single* model processing a single prompt. The multi-agent setting determines how adversarial context enters the context window, but the feature-level hijacking occurs identically regardless of provenance. Any deployment where an LLM processes mixed-trust context is susceptible: retrieval-augmented generation (RAG) over untrusted corpora, customer service agents evaluating user-submitted evidence, code assistants processing user-provided context, or clinical decision support systems incorporating patient-reported information. In each case, an adversary can selectively frame true statements to suppress the model's task features while its awareness features—and behavioral outputs—appear unaffected. This generality follows directly from our cross-domain replication (Table 3): the suppression mechanism recruits domain-specific task features but operates through a shared attention-reallocation pathway that is architecture-level, not domain-level.

Deployed systems that pass behavioral evaluations may nonetheless have compromised internal representations, making feature-level monitoring essential for high-stakes applications. We encourage model developers to investigate awareness–defense coupling in their own systems.

Acknowledgments

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A Feature Identification Details

SAE Configuration. We use GemmaScope 2 16K JumpReLU residual-stream SAEs. For each model size, we analyze features at two layers selected to span mid-network and late-network representations: 4B (L17, L22), 12B (L31, L41), 27B (L31, L40).

Feature Selection Protocol. Task features are identified as those with the highest activation differential (neutral $>$ chaos) that also correlate with correct-branch output probability ($r > 0.8$). Awareness features are identified as those with the highest activation differential (chaos $>$ neutral) that activate specifically in the presence of manipulative framing. We select the top 3 features in each category for ablation experiments.

27B-IT Feature Details. Task features at L40: IDs $\{22, 296, 14680\}$, neutral mean activation 722.5, chaos mean 98.6 (86.3% suppression). Awareness features at L40: IDs $\{2119, 11843, 2145\}$, neutral mean 147.7, chaos mean 442.6 ($3.0\times$ boost). For comparison, 27B-PT awareness features show a $10.4\times$ boost under the same conditions.

B Ablation Method Details

Feature Swap. During chaos-condition generation, we zero the activations of the top-3 awareness features and measure change in task-feature activation and output probability. Recovery is defined as the fraction of the neutral–chaos task-feature gap that is closed.

Attention Knockout. We zero the attention output (value-weighted attention) at all positions corresponding to chaos-message tokens. This directly blocks the hypothesized hijacking pathway.

Activation Patching. For each layer ℓ, we replace the chaos-condition residual stream at layer ℓ with the neutral-condition residual stream at the same layer, and propagate forward. We report the recovery at the layer achieving maximum recovery.

C Full Recovery Prompt Results

Recovery prompts (L1–L5) are textual interventions—not internal representation probes—that provide progressively richer contextual information to the model after hijacking:

- **L1**: Minimal hint (single keyword).
- **L2**: Contextual restatement of the problem.
- **L3**: Explicit identification of the correct branch.
- **L4**: Warning that the previous context was manipulative.
- **L5**: Full neutral-condition context appended.

The anomalous L3 recovery in 27B-IT (30.4% vs. $< 5\%$ at other levels) is consistent with L3 providing the correct answer *directly*, bypassing the awareness–defense coupling that is severed in the IT model. The PT model recovers substantially on L2 (53.5%) and L5 (34.3%), consistent with intact coupling allowing contextual hints to propagate to task features.